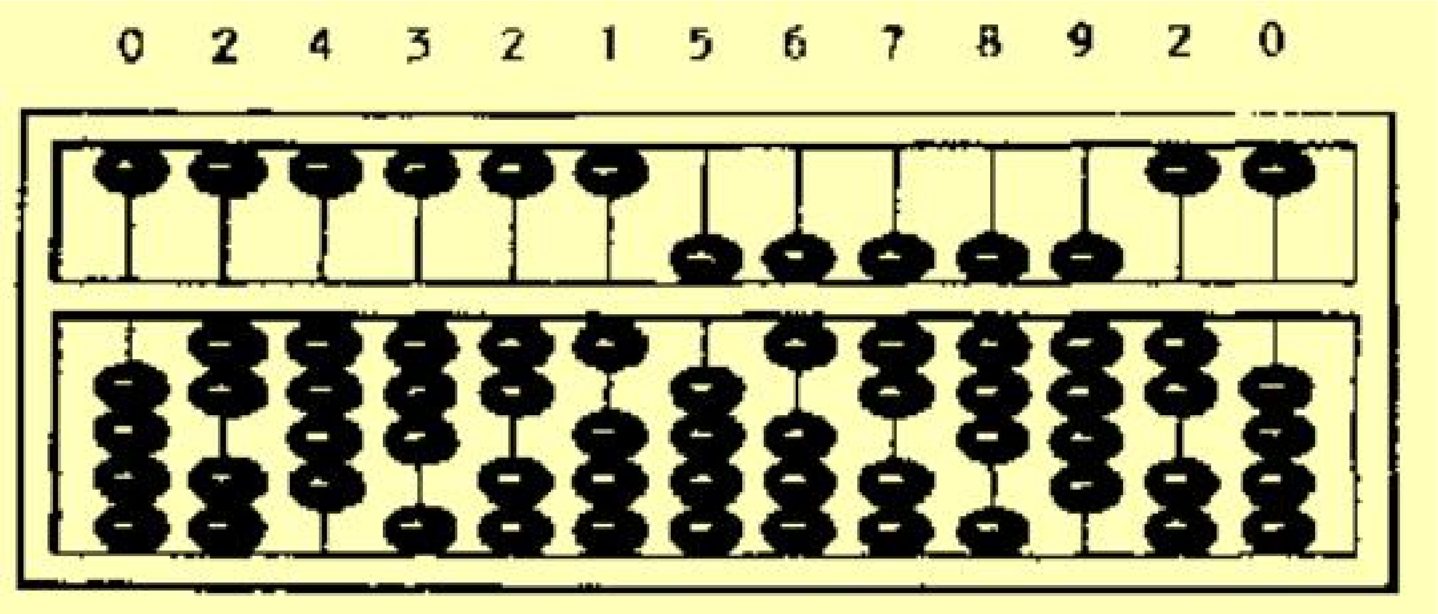
History and Evolution of Computer

Lecture 4

Evolution of Computers

The abacus is a manually operated digital computer used in ancient civilizations

The use of the word abacus dates before 1387 AD



Computer Generations

"Generation" in computer talk is a step in technology. It provides a framework for the growth of computer industry

Originally it was used to distinguish between various hardware technologies, but now it has been extended to include both hardware and software

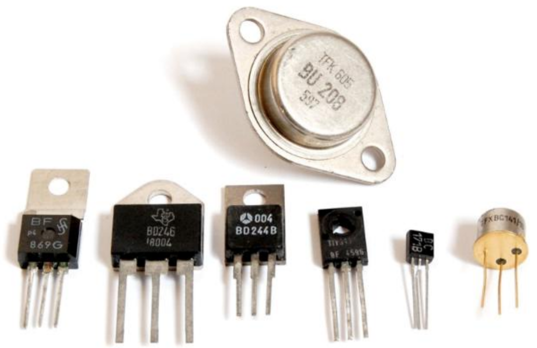
Till today, there are five computer generations

1st Generation Computer

Generation (Period)	Key hardware technologies	Key software technologies	Key characteristics	Some representative systems
First (1942-1955)	Vacuum tubes Electromagnetic relay memory Punched cards secondary storage	Machine and assembly languages Stored program concept Mostly scientific applications	Bulky in size Highly unreliable Limited commercial use and costly Difficult commercial production Difficult to use	ENIAC EDVAC EDSAC UNIVAC I IBM 701

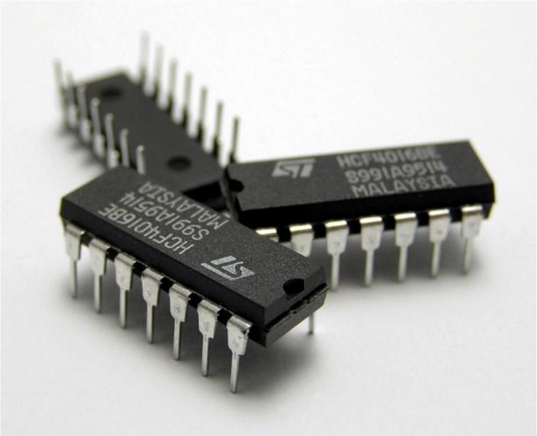
2nd Generation Computer

Generation (Period)	Key hardware technologies	Key software technologies	Key characteristics	Some representative systems
Second (1955-1964)	Transistors Magnetic cores memory Magnetic tapes Disks for secondary storage	Batch operating system High-level programming languages Scientific and commercial applications	Faster, smaller, more reliable and easier to program than previous generation systems Commercial production was still difficult and costly	Honeywell 400 IBM 7030 CDC 1604 UNIVAC LARC



3rd Generation Computer

Generation (Period)	Key hardware technologies	Key software technologies	Key characteristics	Some rep. systems
Third (1964-1975)	ICs with SSI and MSI technologies Larger magnetic cores memory Larger capacity disks and magnetic tapes secondary storage Minicomputers; upward compatible family of computers	Timesharing operating system Standardization of high-level programming languages Unbundling of software from hardware	Faster, smaller, more reliable, easier and cheaper to produce Commercially, easier to use, and easier to upgrade than previous generation systems Scientific, commercial and interactive on-line applications	IBM 360/370 PDP-8 PDP-11 CDC 6600



4th Generation Computer

Generation (Period)	Key hardware Technologies	Key software technologies	Key characteristics	Some rep. systems
Fourth (1975-1989)	ICs with VLSI technology Microprocessors; semiconductor memory Larger capacity hard disks as in-built secondary storage Magnetic tapes and floppy disks as portable storage media Personal computers Supercomputers based on parallel vector processing and symmetric multiprocessing technologies Spread of high-speed computer networks	Operating systems for PCs with GUI and multiple windows on a single terminal screen Multiprocessing OS with concurrent programming languages UNIX operating system with C programming language Object-oriented design and programming PC, Network-based, and supercomputing applications	Small, affordable, reliable, and easy to use PCs More powerful and reliable mainframe systems and supercomputers Totally general purpose machines Easier to produce commercially Easier to upgrade Rapid software development possible	IBM PC and its clones Apple II TRS-80 VAX 9000 CRAY-1 CRAY-2 CRAY-X/MP

5th Generation Computer

Generation (Period)	Key hardware technologies	Key software technologies	Key characteristics	Some rep. systems
Fifth (1989-Present)	ICs with ULSI technology Larger capacity main memory, hard disks with RAID support Optical disks as portable read-only storage media Notebooks, powerful desktop PCs and workstations Powerful servers, supercomputers Internet Cluster computing	Micro-kernel based, multithreading, distributed OS Parallel programming libraries like MPI & PVM JAVA World Wide Web Multimedia, Internet applications More complex supercomputing applications	Portable computers Powerful, cheaper, reliable, and easier to use desktop machines Powerful supercomputers High uptime due to hot-pluggable components Totally general purpose machines Easier to produce commercially, easier to upgrade Rapid software development possible	IBM notebooks Pentium PCs SUN Workstations IBM SP/2 SGI Origin 2000 PARAM 10000